

SEMESTER IV

1	Name of Course/Module : TECHNOPRENEURSHIP																
2	Course Code: TCP 241																
3	Name(s) of academic staff:																
4	Rationale for the inclusion of the course /module in the programme: The course is designed to explore the entrepreneurial mind set and culture utilizing a technology or engineering background. It provides an introduction to the rapidly evolving technical world of creative new ventures that utilizes internet marketing.																
5	Semester and Year offered: year 2 semester 4																
6	Course Hours	Face to Face				ILT	TSLT										
		L	T	P	O												
	L=Lecture T=Tutorial P=Practical O=Others TSLT=Total student learning time	49	32	-	6	81	168										
7	Credit Value:3																
8	Prerequisite: Nil																
9	Course Learning Outcomes: On completion of this course students will be able to: - Describe the entrepreneurial process from the generation of creative ideas. - Analyse the relevant information of market needs and provide a solution to a key problem. - Identify the various challenges in technopreneurship.																
10	Transferable Skills: - Critical Thinking & Problem Solving Skills - Information Management & Life Long Learning - Evaluating results																
11	Teaching –learning and assessment strategy - Lectures - Tutorials At the end of the programme, students are given an opportunity to evaluate the course and the lecturer.																
12	Synopsis: This course intends to give an understanding of Technopreneurship fundamentals which will further encourage science and engineering students to explore the entrepreneurial and R&D career paths. It covers the topics like introduction to entrepreneurship, idea creation, innovation & creativity and product identification.																
13	Mode of Delivery: Lectures, Tutorials.																
14	Assessments Methods and Types: <table><tr><td>Assignments</td><td>20%</td></tr><tr><td>Mid Exam</td><td>20%</td></tr><tr><td>Final Exam</td><td>50%</td></tr><tr><td>Quiz</td><td>10%</td></tr><tr><td>Total</td><td>100%</td></tr></table>							Assignments	20%	Mid Exam	20%	Final Exam	50%	Quiz	10%	Total	100%
Assignments	20%																
Mid Exam	20%																
Final Exam	50%																
Quiz	10%																
Total	100%																
	Content Outline of the course/module and the SLT per topic																

15	No	Subject description	Face to face				ILT	Total
			Lecture	Tutorial	Practical	Others		
	1.	Introduction: - Basic concept on Technopreneurship - Differential between Entrepreneurship and Technopreneurship - Discuss the traits and characteristics of technopreneurship - Identify the challenges in technopreneurship	3	2	-	-	5	10
	2.	Innovation and Ideas: - What is innovation? - Research vs development – translational research - Types of innovation: product, process, and business model - Innovation-driven vs small-medium enterprise - Organization-driven vs market-driven ideas	4	3	-	-	7	14
	3.	Value Proposition: - Benefits Vs features, relation to needs and high value adding - Solution driven or efficiency improvement - Value=benefits/cost - Value proposition statement including needs, approach, benefits per cost, competition	4	4	-	-	8	16
	4.	Customers: - Customer needs, pain points and demographics - Market research and validation - The decision-making process - Target customer profile, persona	3	3	-	-	6	12
	5.	Competitive Advantage, Markets: - Classes of competitors= - Product differentiation, positioning - Market structures - Market segments, size - Beachhead market and creating your market	5	3	-	-	8	16

	6.	Introduction to Intellectual Property: - What is IP? Why have IP protection? Cost of protection - Copyright, trademarks - Patents, trade secrets, contracts, non-disclosure and non-compete agreements	5	3	-	-	8	16
	7.	Execution and Business Plan: - Roadmap for research, development, and production - Budget and timeline - Sales and marketing plans; cost of customer acquisition, customer lifetime value - Plans for R&D, operations, sales and marketing, human resources - Lean concepts and organization	5	3	-	-	8	16
	8.	Financial Analysis and Accounting Basics: - Cash flow statements and projection - Income (P&L) statements; accrual accounting, depreciation, operating expenses - Balance sheets; equity, liability - Breakeven time	5	3	-	-	8	16
	9.	The Product or Service: - What is the core that makes it special? - Minimum viable product and iterative design - Cost of goods sold - Product development plan, Gantt chart	5	3	-	-	8	16

16.	10.	IT Business Development Models: - Organize Financial plan for new ventures in technopreneurship - Time value of money - Revenue generation - Price structure; price elasticity - Channels of distribution - Strategic partners - Develop IT Business Plan - Define the meaning of a Business Plan - Identify the Importance of a Business - Explain the Criteria of a good Business - Determine business plan outline - Create a IT business plan for a given business	6	2	-	-	8	16
	11.	Ethics and social responsibility: - Ethics. codes of ethics; theoretical frameworks; broader ethical considerations - Social businesses	2	2	-	-	4	8
	12.	Globalization: - Cultural differences in communication - Ethical issues	2	2	-	-	4	8
		Total	49	32	-	-	81	162
16.	Main references supporting the course: - Scott Shane's finding fertile ground: Identifying Extraordinary Opportunities for New Ventures (2005) - Kuratko and Hodgetts' Entrepreneurship: Theory, Process, Practice (6th Ed) - Bagley,C, and C.Dauchy (1998), The Entrepreneur's Guide to Business Law, West - Berndt,S. (1997), The 10 Commandments for Building a Growth Company (3rd Ed), Archipelago Press							

1	Name of Course/Module :WEB DESIGNING					
2	Course Code: WBD-113					
3	Name(s) of academic staff:					
4	Rationale for the inclusion of the course /module in the programme: This course is designed to introduce the student to the tools and facilities of web design: page composition and code validation. It helps students to understand the software technologies together to produce web design projects.					
5	Semester and Year offered: year 2 semester 4					
6	Course Hours	Face to Face				ILT
		L	T	P	O	TSLT
	L=Lecture T=Tutorial P=Practical O=Others TSLT=Total student learning time	51	4	27	6	82
7	Credit Value: 3					
8	Prerequisite: Nil					
9	Course Learning Outcomes: On completion of this course students will be able to: • Create an Information Architecture document for a web site. • Apply an organized approach to trouble shooting and solving problems. • Construct a web site that conforms to the web standards of today and includes e-commerce and web marketing useful for own venture.					
10	Transferable Skills: • Critical Thinking and problem solving skills • Ethics, moral and professionalism • Information management and lifelong learning					
11	Teaching –learning and assessment strategy • Lectures • Tutorials At the end of the programme, students are given an opportunity to evaluate the course and the lecturer.					
12	Synopsis: This course is designed to introduce the realm of web design so as to make understand how HTML works, and proceed to more advanced and complicated structures and concepts of web design, such as CSS and layout control.					
13	Mode of Delivery: Lectures, Tutorials, practical.					

14	Assessments Methods and Types: Assignments 20% Mid Exam 20% Final Exam 50% Quiz 10% Total 100%							
15	Content Outline of the course/module and the SLT per topic							
	No	Subject description	Face to face				ILT	Total
			Lecture	Tutorial	Practical	Others		
	1.	An Overview of the internet and the web: • Evolution, Advantage and Disadvantage of Internet • Intranet , Extranet and Intranet • DNS and IP • URL • Internet Services • WWW • Web Pages Static and Dynamic Web Pages • Web Browser • Search Engine	6	2	-	-	8	16
	2.	Website Development Process: • Introduction • Gathering Information • Planning Design • Content Writing and Assembly • Coding • Testing, Review, and Launch • Maintenance	3	-	2	-	5	10
3.	Website Structure Design: • Website Structure • Why Does Your Website's Structure Matter? • Structuring Your Website • Plan Before You Do, • Symmetry Is Key • Pay Attention to Page Slugs • Make Your Menu Front • Center • Keep It Simple	6	-	3	-	9	18	

	4	Website Design Principle: - Purpose Communication - Typefaces - Colours - Images - Navigation - Grid based layouts - Load time - Mobile friendly	4	2	-	-	6	12
	5.	Webpage Layout: - Introduction - The Zig-Zag Layout - The F Layout - Full Screen Photo - One-Column Layout - Featured Image Layout - Asymmetrical Layout - Split Screen Layout - Headline and Thumbnails - Gallery Layout Modular - Card Layout / Block Layout - Magazine Layout - Single Page Layout - Radial Symmetry Layout	9	-	4	-	13	26
	6.	Color and Graphics on the web: - Use of color in web - Use of different fonts in web - Use of Graphics element - Image Tag	2	-	2	-	4	8
	7.	Audio and Video on web: - Use of Audio and Video in WEB - implementation of Audio and video tags with its attributes	2	-	2	-	4	8
	8.	User Interface Design: - Introduction to UI - Introduction to UX - Different tools for wireframe design - Creating UI from wireframe	2	-	2	-	4	8

16.	9.	Style Sheet and Controlling Page: - Using inline style in HTML - CSS and its uses - CSS for responsive design	2	-	2	-	4	8
	10.	Form Design: - Form in Web - INPUT tag of HTML and its uses in form - Type of input field in form	2	-	2	-	4	8
	11.	Website Accessibility: - Accessibility in Context - What is Web Accessibility - Accessibility is Important for Individuals, Businesses, Society Making the Web - Accessible Evaluation - Accessibility Examples	5	-	3	-	8	16
	12.	Website Testing Introduction: - Functionality Testing - Usability testing - Interface testing - Compatibility testing - Performance testing - Security Testing - Cross Browser Testing	5	-	3	-	8	16
	13.	Website Usability Evaluation: - Introduction - FOCUS GROUPS CARD - SORTING FIRST CLICK - TESTING RECRUITING - PARTICIPANTS	3	-	2	-	5	10
		Total	51	4	27	-	82	164
16.	Main references supporting the course: Building Websites All-in-One for Dummies by David Karlins, Doug Sahlin 2019							

1	Name of Course/Module : WIRELESS COMMUNICATION																
2	Course Code: WLC- 234																
3	Name(s) of academic staff:																
4	Rationale for the inclusion of the course /module in the programme: The course provides an introduction to wireless communication systems. It provides a comprehensive overview and advanced knowledge of modern mobile and wireless communication systems.																
5	Semester and Year offered: year 2 semester 4																
6	Course Hours	Face to Face				ILT	TSLT										
		L	T	P	O												
	L=Lecture T=Tutorial P=Practical O=Others TSLT=Total student learning time	46	14	-	6	60	126										
7	Credit Value:4																
8	Prerequisite: Nil																
9	Course Learning Outcomes: On completion of this course students will be able to: - Analyse and design receiver and transmitter diversity techniques. - Design wireless communication systems with key 3G (e.g., CDMA) and 4G (OFDM) technologies. - Describe and differentiate four generations of wireless standard for cellular networks.																
10	Transferable Skills: - Critical Thinking & Problem Solving Skills - Information Management & Life Long Learning - Evaluating results																
11	Teaching –learning and assessment strategy - Lectures - Tutorials At the end of the programme, students are given an opportunity to evaluate the course and the lecturer.																
12	Synopsis: The course addresses the fundamentals of wireless communications and provides an overview of existing and emerging wireless communications networks.It helps students to understand the different techniques used to support voice, data and video communication in wireless systems.																
13	Mode of Delivery: Lectures, Tutorials.																
14	Assessments Methods and Types: <table><tr><td>Assignments</td><td>20%</td></tr><tr><td>Mid Exam</td><td>20%</td></tr><tr><td>Final Exam</td><td>50%</td></tr><tr><td>Quiz</td><td>10%</td></tr><tr><td>Total</td><td>100%</td></tr></table>							Assignments	20%	Mid Exam	20%	Final Exam	50%	Quiz	10%	Total	100%
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	Content Outline of the course/module and the SLT per topic																

15	No	Subject description	Face to face				ILT	Total
			Lecture	Tutorial	Practical	Others		
	1.	Introduction to wireless communication systems: • Evolution of mobile radio • Communications Examples of wireless comm. • Systems Paging systems • Cordless telephone systems • Comparison of various wireless systems	7	2	-	-	9	18
	2.	Modern Wireless Communication Systems: • Second generation cellular networks • Third and fourth generation wireless networks • Wireless in local loop • Wireless local area networks • Blue tooth and Personal Area	7	2	-	-	9	18
	3.	Introduction to Cellular Mobile Systems: • Spectrum Allocation • Basic Cellular Systems • Performance Criteria • Operation of cellular systems • Analogy cellular systems • Digital Cellular Systems	6	2	-	-	8	16
	4.	Cellular System Design Fundamentals: • Frequency Reuse • Channel assignment strategies • Handoff Strategies, • Interference and system capacity, • Tracking and grade off service • Improving coverage and capacity	6	2	-	-	8	16

16.	5.	Multiple Access Techniques For Wireless Communication: • Introduction to Multiple Access • FDMA TDMA, • Spectrum Multiple Access • Space division multiple access • Packet ratio • Capacity of a cellular systems	6	2	-	-	8	16
	6.	Wireless Networking: • Difference between wireless and Fixed telephone networks • Development of wireless networks • Fixed network transmission • Hierarchy Traffic routing in wireless networks, Wireless data services Common • Channel signaling, ISDN (Integrated Services digital Networks) • Advanced intelligent networks.	9	2	-	-	11	22
	7.	Intelligent Cell Concept And Application: • Intelligent cell concept applications of intelligent micro-cell Systems In-Building Communication • CDMA cellular	5	2	-	-	7	14
		Total	46	14	-	-	60	120
16.	Main references supporting the course: • Wireless Communication 1st Edition by Andrea Goldsmith							

1	Name of Course/Module : ETHICAL HACKING																
2	Course Code: EHG-235																
3	Name(s) of academic staff:																
4	Rationale for the inclusion of the course /module in the programme: The course module enables students to learn the basic concepts and introduction of ethical hacking with the overview of current security trends.																
5	Semester and Year offered: year 2 semester 4																
6	Course Hours	Face to Face				ILT	TSLT										
		L	T	P	O												
	L=Lecture T=Tutorial P=Practical O=Others TSLT=Total student learning time	34	2	24	6	60	126										
7	Credit Value:4																
8	Prerequisite: Nil																
9	Course Learning Outcomes: On completion of this course students will be able to: - Describe the overview of vulnerability research and vulnerability classification. - Evaluate the security status of systems and suggest solutions for removing security vulnerabilities. - Obtain information on the legal aspects of performing ethical hacking and to judge what is within and outside permitted activities.																
10	Transferable Skills: - Critical Thinking & Problem Solving Skills - Information Management & Life Long Learning - Evaluating results																
11	Teaching –learning and assessment strategy - Lectures - Tutorials At the end of the programme, students are given an opportunity to evaluate the course and the lecturer.																
12	Synopsis: The main focus of this course is to provide in depth knowledge on how to defend against various sniffing techniques. It helps students to be able to perform practical penetration testing using up-to-date tools and techniques. It covers foot printing and reconnaissance, scanning networks, enumeration and system hacking.																
13	Mode of Delivery: Lectures, Tutorials, practical.																
14	Assessments Methods and Types: <table><tr><td>Assignments</td><td>20%</td></tr><tr><td>Mid Exam</td><td>20%</td></tr><tr><td>Final Exam</td><td>50%</td></tr><tr><td>Quiz</td><td>10%</td></tr><tr><td>Total</td><td>100%</td></tr></table>							Assignments	20%	Mid Exam	20%	Final Exam	50%	Quiz	10%	Total	100%
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Quiz	10%																
Total	100%																

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Content Outline of the course/module and the SLT per topic							
No	Subject description	Face to face				ILT	Total
		Lecture	Tutorial	Practical	Others		
1.	Introduction to Ethical Hacking: • Overview of Current Security Trends • Elements of Information Security • Information Security Threats and Attack Vectors • Hacking Concepts, Types and Phases • Ethical Hacking Concepts and Scope • Information Security Controls • Penetration Testing • Information Security Acts and Laws	3	1	-	-	4	8
2.	Foot printing and Reconnaissance: • Foot Printing Concepts • Foot Printing through Search Engines and Advanced Google Hacking Techniques • Foot Printing through Web Services and Social Networking Sites • Website Foot Printing, Email Foot Printing and Competitive Intelligence • WHOIS, DNS and Network Foot Printing • Foot Printing through Social Engineering • Different Foot Printing Tools and Countermeasures • Foot Printing Penetration Testing	4	-	2	-	6	12
3.	Scanning Networks: • Network Scanning Concepts • Various Scanning Tools • Various Scanning Techniques • Various Techniques to Scan Beyond IDS and Firewall • Banner Grabbing • Drawing Network Diagrams • Overview of Scanning Pen Testing	2	-	2	-	4	8

	4.	Enumeration: - Enumeration Concepts - Different Techniques for NetBIOS Enumeration - Different Techniques for SNMP Enumeration - Different Techniques for LDAP and NTP Enumeration - Different Techniques for SMTP and DNS Enumeration - Enumeration such as IPSec, VoIP, RPC and Linux/Unix Enumeration - Different Enumeration Countermeasures - Overview of Enumeration Pen Testing	3	1	-	-	4	8
	5.	Vulnerability Analysis: - Vulnerability Research and Classification - Vulnerability Assessment - Vulnerability Management Life Cycle - Different Approaches of Vulnerability Assessment Solutions - Different Types of Vulnerability Assessment Tools	2	-	2	-	4	8
	6.	System Hacking: - CEH Hacking Methodology - Techniques to Gain Access to the system - Privilege escalation techniques - System hacking Penetration Testing	1	-	1	-	2	4
	7.	Malware Threats: - Malware and Malware Propagation Techniques - Overview of Trojans, their types and how they infect systems - Viruses, their types and how they infect files - Overview of computer worms	2	-	2	-	4	8

	8.	Sniffing: - Sniffing techniques - How to defend against various sniffing techniques - Different techniques to detect sniffing	1	-	1	-	2	4
	9.	Social Engineering: - Social Engineering concepts - Various Social Engineering Techniques - Insider threats - Identity theft concepts - Social engineering countermeasures	1	-	1	-	2	4
	10.	Denial of Service - DoS/DDoS attack techniques - Botnet Network - DoS and DDoS attack tools - Techniques to detect DoS and DDoS attacks	1	-	1	-	2	4
	11.	Session Hijacking: - Application level session hijacking - Network level session hijacking - Session hijacking tools - Session hijacking countermeasures	2	-	1	-	3	6
	12.	Evading IDS, Firewalls and Honeypots: - IDS, Firewall and Honeypot Solutions - Techniques to Bypass IDS - Techniques to Bypass Firewalls - IDS/ Firewall evading tools - Techniques to detect honeypots	1	-	2	-	3	6
	13.	Hacking Web Servers: - Web Server Attacks - Web Server Attack Methodology - Web server attack countermeasures - Patch Management Concepts - Web server security tools	2	-	1	-	3	6

	14.	Hacking Web Applications: - Web Application concepts - Web Application threats - Web Application Hacking Methodology - Web Application Hacking Tools	2	-	1	-	3	6
	15.	SQL Injection: - Types of SQL Injection Attacks - SQL Injection Methodology - SQL Injection Tools	1	-	1	-	3	6
	16.	Hacking Wireless Networks: - Wireless Encryption Algorithms - Wireless threats - Wireless hacking methodology - Wireless hacking tools	1	-	1	-	4	8
	17.	Hacking Mobile Platforms: - Mobile Platform Attack Vectors - Various Android Threats and Attacks - Various iOS Threats and Attacks - Mobile security guidelines and security tools	2	-	2	-	4	8
	18.	IoT Hacking: - IoT Concepts - IoT Threats and Attacks - IoT Hacking Countermeasures - IoT Security Tools	1	-	1	-	2	4
	19.	Cloud Computing: - Cloud Computing Threats - Cloud Computing Attacks - Cloud Computing Security	1	-	1	-	2	4
	20.	Cryptography: - Cryptography Algorithms - Cryptography Tools - Email Encryption - Cryptanalysis Tools	1	-	1	-	2	4
		Total	34	2	24	-	60	120
16.	Main references supporting the course: Ethical Hacking: A Hands-on Introduction to Breaking In by Daniel Graham							

1	Name of Course/Module : DATABASE MANAGEMENT SYSTEM						
2	Course Code: DMS-236						
3	Name(s) of academic staff:						
4	Rationale for the inclusion of the course /module in the programme: This course offers the knowledge on fundamental elements of relational database management systems as well as importance of data & data management.						
5	Semester and Year offered: year 2 semester 4						
6	Course Hours	Face to Face				ILT	TSLT
		L	T	P	O		
	L=Lecture T=Tutorial P=Practical O=Others TSLT=Total student learning time	45	11	24	6	80	166
7	Credit Value: 4						
8	Prerequisite: Nil						
9	Course Learning Outcomes: On completion of this course students will be able to: • Introduce the fundamental concepts and methods necessary for the design and use of a database system. • Demonstrate practical experience in applying these concepts and methods using commercial database management systems. • Identify the Advantages & Disadvantages of E-R Data Model.						
10	Transferable Skills: • Critical Thinking & Problem Solving Skills • Information Management & Life Long Learning • Evaluating results						
11	Teaching –learning and assessment strategy • Lectures • Tutorials At the end of the programme, students are given an opportunity to evaluate the course and the lecturer.						
12	Synopsis: This course introduces the fundamentals of database technology. Topics covered include: database concepts, Database System Architecture, E-R model, relational model, database design theory, database languages, transaction management, concurrency control and database recovery.						
13	Mode of Delivery: Lectures, Tutorials, Practical.						
14	Assessments Methods and Types: Assignments 20% Mid Exam 20% Final Exam 50% Quiz 10% Total 100%						

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Content Outline of the course/module and the SLT per topic							
No	Subject description	Face to face				ILT	Total
		Lecture	Tutorial	Practical	Others		
1.	Introduction to the Database Systems: • Importance of Data & Data Management • How Data are Stored in Database • Different Database Systems • Physical & Logical Structure of Database • Database Management Systems and Database Systems • Database Architecture • Difference Between Distributed Database & Relational Database • History	6	3	-	-	9	18
2.	The Relational Data Models: • Entity Relationship and Object • ER Diagram • Importance of ER Diagram • Relational Model • Advantages and Disadvantages of E-R Data Model	3	-	2	-	5	10
3.	Relational Algebra &Calculus: • The fundamental Operations of Relational Algebra Operators: Select, Project, Rename, Union, Intersection, Minus, Cartesian Product, Theta Join, Equijoin, Natural Join, Division • Unary & Binary Operator • Project • Union (Union, Union All, Intersect) • Set Different (Minus Operation) • Cartesian Product	5	-	3	-	8	16

	4.	SQL: • Sql Introduction • Types of Sql (DDL, DML, DCL) • Execution Process of Sql • Execution Process of Sql • Sql Fundamentals: Multi table Queries • Duplicate Rows • Row Selection • Search Conditions • The Comparison Test (=, <, >, <=, >=) • The Range Test (BETWEEN) • The Set Membership Test (IN) • The Pattern Matching Test (LIKE) • The Null Value Test (IS NULL) • Compound Search Conditions (AND, OR and NOT) • Sorting Query Results (ORDER BY Clause) • Developing Sub Queries	5	-	10	-	15	30
	5.	Conceptual Design: • Conceptual Design Process • Requirement Analysis • Identify the Relation	4	-	2	-	6	12
	6.	Logical Design: • Logical Design Process • Design Entity Relation Diagram • Create Tables and Constraint • Create Referential Keys	4	-	3	-	7	14
	7.	Normalization: • Use of Normalization • Different form of Normalization	2	-	2	-	4	8
	8.	Database Technology: • Different Database Technologies • Database Client & Server Processing • OLAP & OLTP Database Techniques	3	2	-	-	5	10

	9.	Distributed Architecture: - Distributed Database Architecture - Advantages and Disadvantage of Distributed Database	3	2	-	-	5	10
	10.	Data Evaluation and Transaction: - Transaction Management - ACID Properties - Database Evaluation Process	4	2	-	-	6	12
	11.	Data Analysis: - Data Analysis Process - Types of Data Analysis - Data Analysis Steps	4	2	-	-	6	12
	12.	Database and the World Wide Web: - Web Data Management - Web Search - Web Crawling	2	-	2	-	4	8
		Total	45	11	24	-	80	160
16.	Main references supporting the course: - Database System Concepts by Abraham Silberschatz and S Sudarshan. - Database Management Systems by Raghu Ramakrishnan							